The Adaptive Agent

Energy markets, part 2 — an agent that rewrites its strategy

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Vector Institute | July 2026

What we'll cover

From analyst to learner

Yesterday's Analyst Agent with exactly one new box — a mutable strategy file it reads before every forecast and rewrites after.

The mechanism

Four evidence layers, five typed write-tools, and an in-code gate that decides when a finding is allowed to change the forecast.

What it discovered

A real finding from its 2025 curriculum, then a +5% before/after on a protected 2026 window — the Strait of Hormuz shock.

One architectural addition turns a capable forecaster into a learner.

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One addition, not a new agent

ANALYST AGENT

Strategy fixed at config time

Instructions never change between runs

Same capable analyst, every single run

ADAPTIVE AGENT

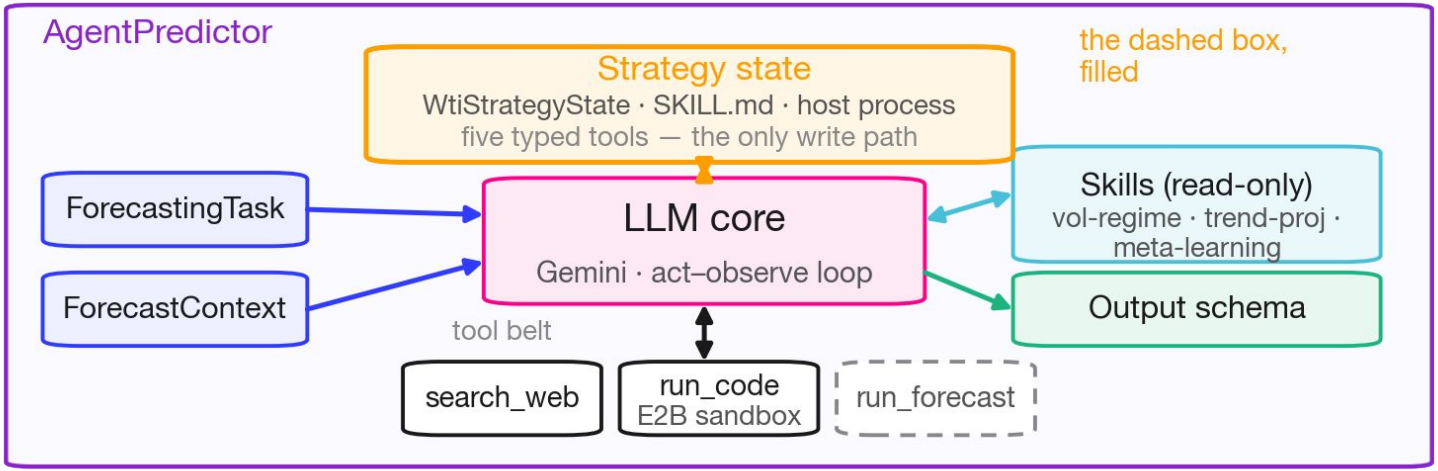
Strategy is a mutable skill file

Updates from its own forecasting experience

Same analyst — now carrying a memory

Identity, tools, output schema, Predictor interface — all identical. One box changes.

The architecture, filled in



The dashed slot from d1-04, now solid: five typed tools, the only write path.

It accumulates experience, not just context.

Observations, hypotheses, and corrections carry from session to session — a living document the agent reads at every prediction.

> The strategy is the memory.

THE MECHANISM

Structured memory, governed

What the agent can update — and the evidence each layer demands

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Four layers, four evidence bars

Layer	Mutation tool	Evidence bar to update it
Observations	record_observation	A pattern that holds across ≥2 forecasts
Hypotheses	open_hypothesis	One strong, durable-looking observation
Calibrations	graduate_hypothesis	Confirmations ≥ threshold (enforced in code)
Approach narrative	update_approach_narrative	A structural shift, not a single result

Bottom-up: you can't reach a calibration without the observation and hypothesis that earn it.

The in-code accept/reject gate

```python
# graduate_hypothesis enforces the gate
if confirms < threshold:
    short = threshold - confirms
    return (
        f"Cannot graduate {hyp_id}: "
        f"have {confirms}, "
        f"need {threshold}. "
        f"Record {short} more."
    )
```

adaptive_agent/skill_tools.py:313

The gate is in code.

The agent can't talk its way past the threshold (set to 2).

- Returns the exact shortfall
- Forces more evidence first
- Not a prompt — a hard rule

CURRICULUM

What it discovered

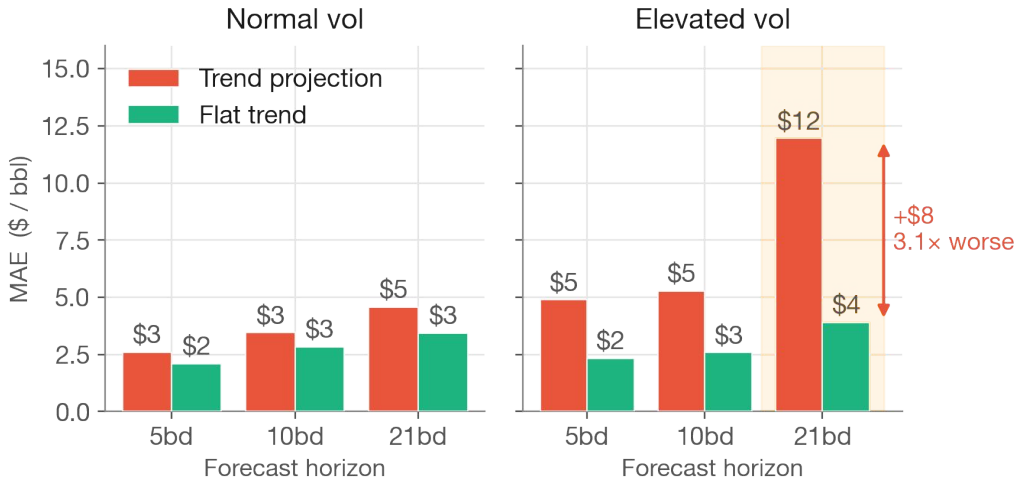
NB05 self-directed study on 2025 → NB06 protected 2026 eval

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The curriculum — NB05

1. **Load 2025 context**
   52 weekly news summaries (OPEC+, geopolitics) + WTI price history up to each week
2. **Self-directed study**
   Runs its own backtests via the vol-regime and trend-projection skills; hunts systematic error
3. **Record findings**
   Logs observations linked to evidence and opens hypotheses — through the typed tools only
4. **Attempt to graduate**
   Calls graduate_hypothesis; the in-code confirmation threshold accepts or rejects the update
5. **Protected eval (NB06)**
   Strategy is frozen, then scored on 8 unseen 2026 origins from Feb–Mar

What the agent found



The agent's own 2025 WTI backtest — logged as hyp-001.

Trend breaks in high vol.

In elevated vol the linear trend blows out — 3× worse than simply holding the last price.

$11.95

trend MAE · 21bd

$3.91

flat-trend MAE · 21bd

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Before and after the curriculum

SEED STRATEGY

Horizon-based approach priors

No observations recorded

No hypotheses, no corrections

TRAINED STRATEGY

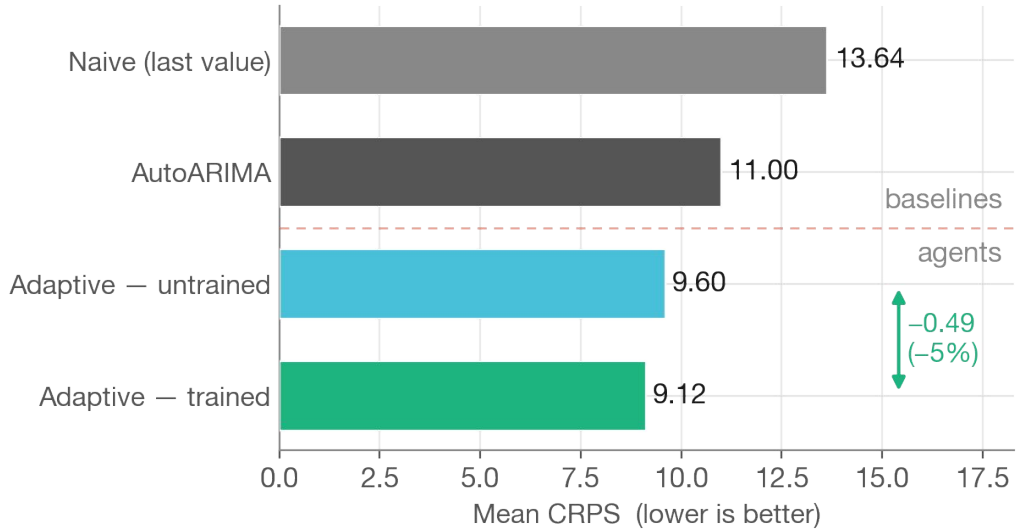
Approach narrative: unchanged

Obs: flat-trend MAE 3× lower at 21bd

hyp-001: open, linked to the evidence

One session, one concrete finding — recorded with the actual backtest numbers.

Before/after on the 2026 eval



Lower CRPS is better — 8 protected 2026 origins it never saw.

Better on the hardest window.

On 8 unseen 2026 origins, training beats the untrained agent and both baselines.

9.12

trained mean CRPS

–5%

vs untrained (9.60)

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Visible in the live rationale

"To prevent extrapolation error (which would project $68.76 via a linear model), we anchor our point forecast on the AutoARIMA baseline of $62.99 — a flat, stable trend with much lower MAE in backtesting. The 80% interval [$55.66, $70.99] uses split-normal interpolation for the elevated volatility regime."

eval_Agent__trained.json · as_of 2026-02-09

Not a rule we gave it.

It cites the curriculum finding in its live Feb-9-2026 rationale.

- Anchors by strategy, not instruction
- Traceable to the 2025 backtest
- Real numbers, this origin

What we still haven't done

01

No validation gate

unproven

Updates commit without checking held-out improvement.

02

No archive

single point

One live strategy — nothing to fall back on if a turn goes wrong.

03

No learned schema

our design

The four layers are ours — the agent works within them, didn't discover them.

The change is visible. But is 'change' the same as 'improve'?

It changes. Does it improve?

The change is visible and auditable. Whether it's improvement — not drift, not overfitting — is the question the next session answers.

Next: d2-03 — Self-improving Agentic Systems